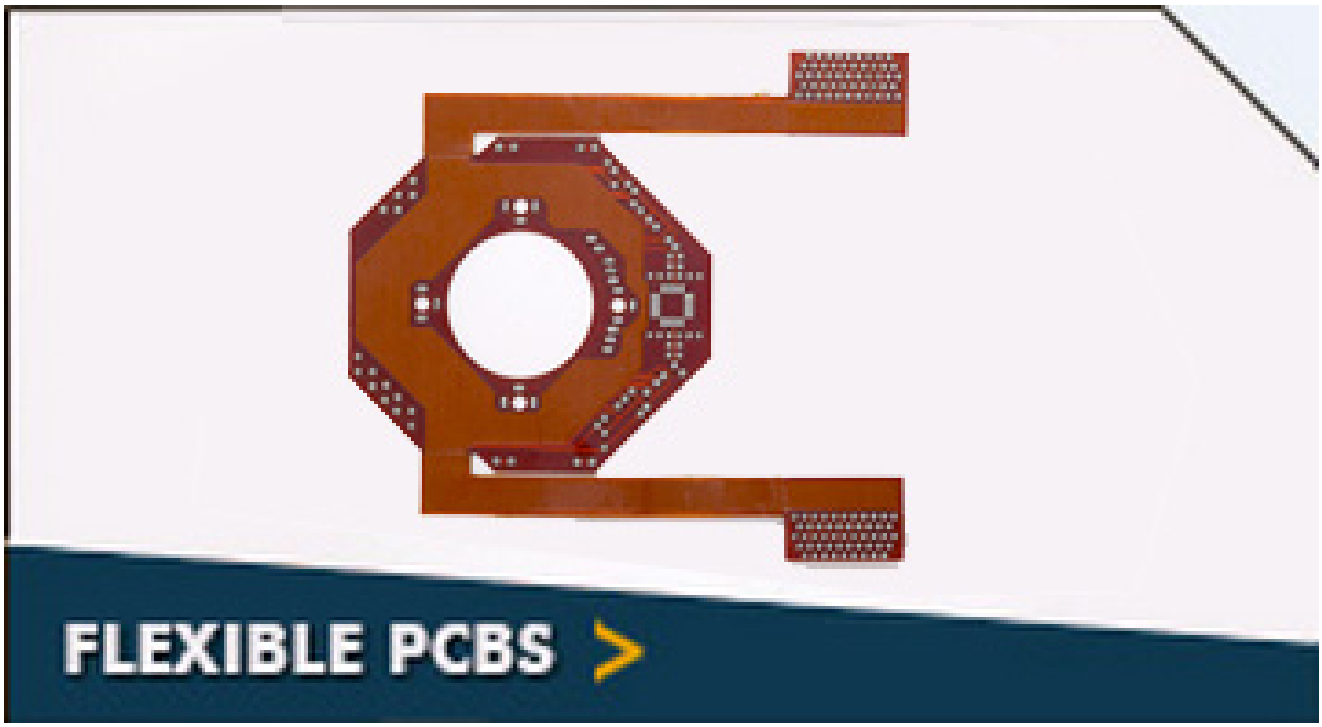
RIGIFLEX TECHNOLOGY, INC.

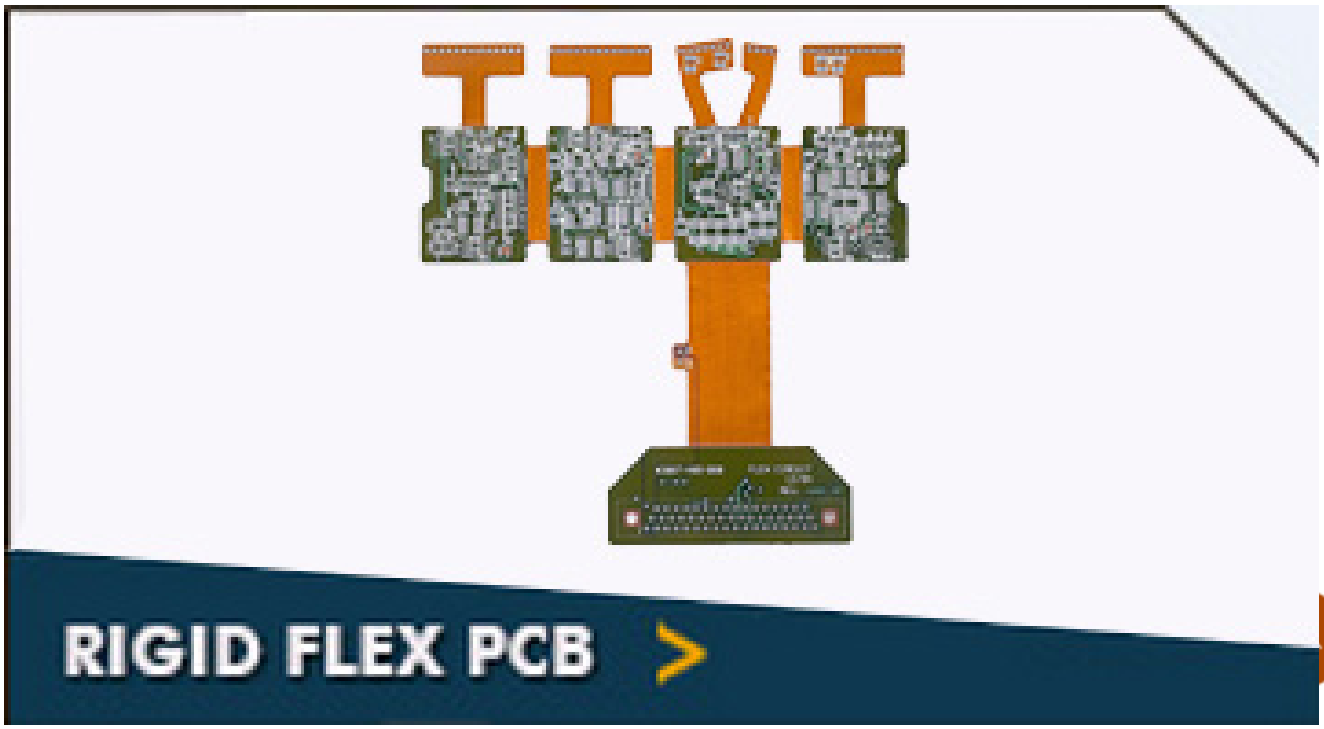
Manufacturers of Rigid, Flex and Rigid-Flex Circuit Boards

(714) 688-1500 (tel:7146881500) pcboards@rigiflex.com (mailto:pcboards@rigiflex.com)

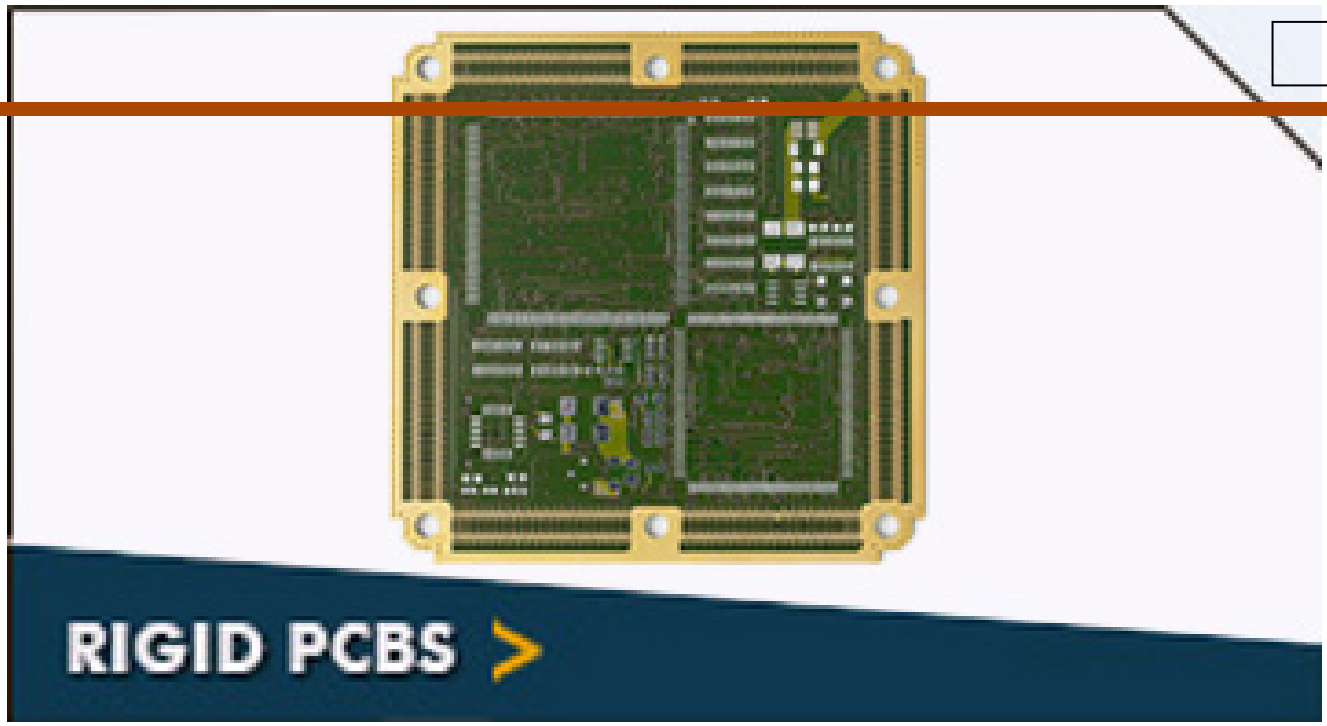
Flex & Rigid-Flex PCBs.. 1-5 Day Quick Turn



(https://www.rigiflex.com/products-services/flexible-pcb/)



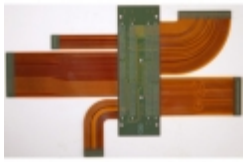
(https://www.rigiflex.com/products-services/rigid-flex-pcb/)



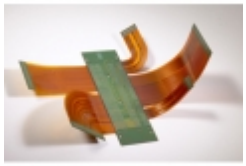
(https://www.rigiflex.com/products-services/rigid-pcb/)

Superior & Trusted Flexible, Rigid-Flex & Rigid PCBs for Over 25 years

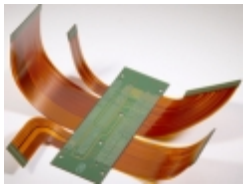
10 Layer Rigid-Flex PCB with 4 Layer Flex Circuit Boards



(https://www.rigiflex.com/wp-content/gallery/10-layer-rigid-flex-with-4-layer-flex/portfolio_group_Rigiflex-006_original_1.jpg?t=1423031887)



(https://www.rigiflex.com/wp-content/gallery/10-layer-rigid-flex-with-4-layer-flex/portfolio_group_Rigiflex-101_original_1.jpg?t=1423031888)



(https://www.rigiflex.com/wp-content/gallery/10-layer-rigid-flex-with-4-layer-flex/portfolio_group_Rigiflex-103_original_1.jpg?t=1423031888)

This is a specially designed 10 layer rigid-flex printed circuit board (PCB), which has 4 additional flexible layers. This provides advantages in terms of stability and flexibility. At Rigiflex, we use Polyimide, Kapton, and PET as the main materials for manufacturing printed circuit boards. Especially for the rigid section, we use specialized Taiyo PSR 4000 white solder masks. This stable, rigid area of the PCB is where the main components are mounted. The supplementary flexible layers allow for a large flex surface, as well as high frequency component to connector connections across the flexible area. Overall, this results in dynamic bending, strong signal integrity, and high reliability.

The 10 layer rigid-flex PCB with 4 layer flex circuit boards that we provide are made from FR4 + PI material. The finish thickness of these printed circuit boards is 1.6 mm. These specially designed PCBs have a decent insulation resistance of about 1000 MW, and an excellent dielectric strength of 5 KV. These circuit boards are surface treated with immersion gold material. However, you can also get the surface finishing done in other materials, such as OSP, and HASL. You can avail these printed circuit boards in a wide range of copper thicknesses, including 0.5 OZ, 1.0 OZ, 2.0 OZ, 3.0 OZ, up to 6 OZ. The board thickness can also be customized from 0.5 mm to 3.0 mm. The selection of solder mask can be done from varied color options, such as white, black, blue, green, red, and Taiyo PSR4000 white.

10 Layer Rigid-Flex PCB with 4 Layer Flex Circuit Boards	
Model:	Rigid Flex PCB
Material:	FR4+PI
Surface treatment:	immersion gold
Width/space:	4/4 mil
Finish thickness:	0.2mm(flex), 1.6mm
Min Hot:	0.1mm

OEM Custom Service	
Board Thickness	0.5mm~3.0mm (0.02″~0.12″)
Copper thickness:	0.5 OZ, 1.0 OZ, 2.0 OZ, 3.0 OZ, up to 6 OZ
Outline:	Routing, punching, V-Cut
Solder mask:	White/Black/Blue/Green/Red, Taiyo PSR4000 white
Legend/Silkscreen Color:	Black/White
Surface finishing:	Immersion Gold, HASL, OSP
Max Panel size:	18″*24″
Packing:	Vacuum/Plastic bag
Layer:	1 – 20 layers

SPECIFICATION

Materials	Polyimide (Kapton)	Polyester (PET)	Remark& Test Method
	1~20(Rigid-Flex & Multilayers FPC)	1~2	
Single Sided	0.050 mm		
Double Sided	0.075 mm		
Drilling P.T.H.	0.2 mm		
Punching	1.0 mm		
Conductor Width (W)	0.025 mm		W0.5mm
Hole Diameter (H)	0.05 mm (withP.T.H.0.1mm)		H1.5mm
Accumulated Pitch (P)	0.05 mm (Special0.03mm)		P25mm
Outline Dimension (L)	0.05 mm		L50 mm
Conductors and Outline (C)	0.15 mm (Special 0.07mm)		C5.0 mm
Conductors and Coverlay	0.3~0.5 mm		
Surface Treatment on	Soft or Hard Ni/Au Sn/Pb (2~60μm) Primary FluxCarbon Printed		
Terminals and land Areas	4~10μm Silver Gel printed		

Insulation Resistance	1000MW		IPC-TM-650 2.6.3.2
			at Ambient
Dielectric Strength	5KV		IPC-TM-650 2.5.6.1
Surface Resistance (W)	5×012	2×015	IPC-TM-650 2.5.17
Volume Resistivity (W-cm)	1×015	1×015	IPC-TM-650 2.5.17
Dielectric Constant (1 MHz)	4	3.4	IPC-TM-650 2.5.5.3
Dissipation Factor (1 MHz)	0.04	0.02	IPC-TM-650 2.5.5.3
Peeling Strength (180 irection)	1.2kgf/cm	1.2kgf/cm	IPC-TM-650 2.4.9
Solder Heat Resistance	260 .C/10secs	210 .C/3secs	
Flammability	94 V-0	94VTM-0	UL94
Water Absorption	2.90%	1.00%	ASTM D570%
			(24 Hours)

Quick Contact

All Fields are Mandatory (*)

Full Name

Email

Telephone

Type Your Message

5465c1

Captcha

Submit



ITAR Registered
M26780

Flex & Rigid Flex
1-5 Day Turn

Over 25 Years
In The Industry

Need it quick? That is our speciality!

FOOTER NAVIGATION

- Home (https://rigiflex.com/)
- Company Profile (https://www.rigiflex.com/company-profile)
- Customers (https://www.rigiflex.com/company-profile/customers)
- Capabilities (https://www.rigiflex.com/capabilities)
- Product & Services (https://www.rigiflex.com/products-services)
- Flexible PCB`s (https://www.rigiflex.com/products-services/flexible-pcb)
- Rigid-Flex PCB`s (https://www.rigiflex.com/products-services/rigid-flex-pcb)
- Rigid PCB`s (https://www.rigiflex.com/products-services/rigid-pcb)
- Quality & Delivery (https://www.rigiflex.com/quality-delivery)
- Industries Served (https://www.rigiflex.com/industry-served)
- Contact (https://www.rigiflex.com/contact-us)
- Privacy Policy (https://www.rigiflex.com/privacy-policy)
- PCB Quote (https://www.rigiflex.com/pcb-quote)
- FAQ (https://www.rigiflex.com/faq)
- Blog (https://rigiflex.com/blog/)
- Sitemap (https://www.rigiflex.com/sitemap)
- Case Study (https://www.rigiflex.com/case-study)
- Certifications (https://www.rigiflex.com/certifications

- XML (https://www.rigiflex.com/sitemap.xml)

FOLLOW US ON

Facebook

(https://www.facebook.com/RigiflexTechnologyInc)

Twitter

(https://twitter.com/rigiflextech)

Rigiflex Technology Inc.

1166 N. Grove Street Anaheim, CA 92806

(714) 688-1500 (tel:7146881500)

(714) 688-1505

pcboards@rigiflex.com (mailto:pcboards@rigiflex.com)

© 2025 RIGIFLEX TECHNOLOGY INC. - EXPERTS IN FLEX AND RIGID-FLEX CIRCUIT BOARDS MANUFACTURING